This paper must be answered in English

INSTRUCTIONS FOR SECTION B

(1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1, 3, 5, 7 and 9.

(2) Refer to the general instructions on the cover of the Question Paper for Section A.

(3) Answer **ALL** questions.

(4) Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.

(5) Graph paper and supplementary answer sheets will be provided on request. Write your Candidate Number, mark the question number box and stick a barcode label on each sheet, and fasten them with string **INSIDE** this Question-Answer Book.

(6) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.

1	**5**
2	**6**
3	**5**
4	**5**
5	**7**
6	**5**
7	**5**
8	**7**
9	**6**
Long Questions Question No.	**Marks**
10	**11**
11	**13**
12	**9**

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(b) The power output of the heater is 2 kW. Estimate the time required. (2 marks)

(c) Explain why the temperature of water remains at the boiling point after the water starts to boil, even when the heater is still on. (1 mark)

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(b) (i) Find the average kinetic energy of the gas molecules. (2 marks)

(ii) The mass of one mole of neon gas is 0.020 kg. Find the root-mean-square speed of the gas molecules. (2 marks)

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(a) Find the tension of the string. (2 marks)

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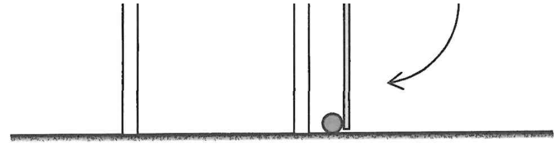

Figure 3.2

(b) (i) If the mass of the ball is 0.05 kg and the collision time is 0.008 s, find the magnitude of the average force acting on the ball by the rod. (2 marks)

(ii) Explain why the momentum of the ball is not conserved before and after the collision. (1 mark)

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Figure 4.1(a)

Figure 4.1(b)

A small sphere is projected along the track with speed u at A. It moves along the track and passes B at $4.0\ \mathrm{m\ s^{-1}}$. Neglect air resistance. ($g = 9.81\ \mathrm{m\ s^{-2}}$)

(a) By using the law of conservation of energy, find u. (2 marks)

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(ii) Estimate the magnitude of the acceleration of the sphere. (1 mark)

(iii) What force provides the centripetal force for the sphere ? (1 mark)

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(b) (i)

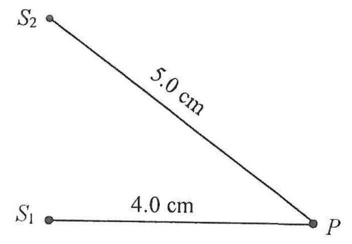

Figure 5.2

P is a point at the water surface as shown in Figure 5.2. Express the path difference at P from S_1 and S_2 in terms of λ. (2 marks)

(ii) Explain briefly what type of interference occurs at P. (2 marks)

(c) If the depth of the water in the ripple tank is increased while other conditions remain unchanged, deduce how the wavelength of the waves will be affected. (2 marks)

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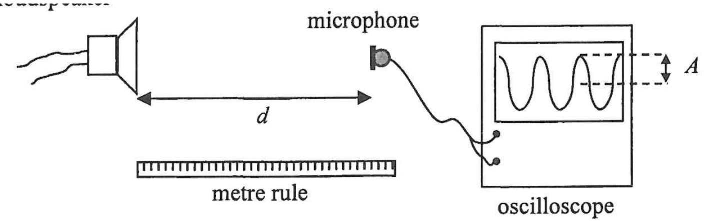

Describe
(I) the procedure of the experiment;

(II) how the data should be analysed using a graphical method to verify whether the hypothesis ($A \propto \frac{1}{d^2}$) is valid.

(5 marks)

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The wallpaper can be easily applied or removed. After removal, the wallpaper does not leave residue or damage the underlying surface. Therefore, the wallpaper is especially suitable for quick updates and temporary setups. Another advantage of the wallpaper is its reusability. It preserves its adhesive properties after being removed and can be reapplied.

(a) Describe how the electrostatic adhesive wallpaper can adhere to a painted wall that is initially neutral. (2 marks)

(b) Explain briefly why the wallpaper preserves its adhesive properties after being removed. (2 marks)

(c) The wallpaper **does not** adhere well to the wall in a humid environment. Explain briefly. (1 mark)

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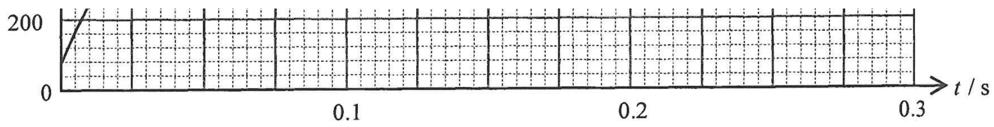

(a) The bulb works at its rated value after R becomes stable. Estimate the rated power of the bulb. (2 marks)

(b) Explain why R initially increases and becomes stable after a period of time. (3 marks)

(c) The filament of the bulb has a length of 0.56 m and a uniform cross-section of diameter 0.022 mm. Find the resistivity of the filament at $t = 0$ s. (2 marks)

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(1 mark)

(b) Estimate the maximum magnetic flux linkage produced by the wire through the search coil. (3 marks)

(c) A student claims that the search coil can be used to measure the resultant magnetic field produced by the Earth and the wire. Explain briefly whether this claim is correct. (2 marks)

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As shown in Figure 10.2, the rain sensor is composed of a light emitting diode (LED) and an infra-red sensor. The LED emits an infra-red signal and the infra-red sensor measures the intensity of the signal. When the intensity changes, the wipers are activated. An infra-red ray from the LED incident at surface A at 45°, and the infra-red ray finally received by the infra-red sensor are drawn in Figure 10.2. Assume surface A is parallel to surface B. It is given that the critical angle of the glass for infra-red is 44.0°.

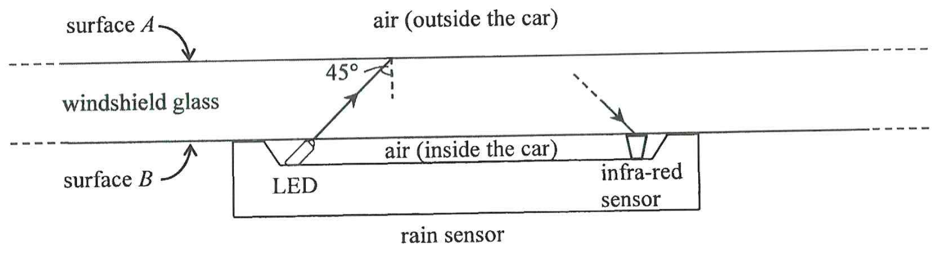

Figure 10.2

(a) Explain how the infra-red ray in Figure 10.2 travels from the LED to the infra-red sensor. (4 marks)

Figure 10.3(a)

Figure 10.3(b)

(i) Given that the refractive index of water for the infra-red is 1.26, find the critical angle of the glass-water interface for the infra-red ray. (3 marks)

(ii) Sketch in Figure 10.3(b) how the infra-red ray propagates after being incident on the glass-water interface. (2 marks)

(iii) The wipers are activated when the intensity of the signal received by the infra-red sensor changes. Explain briefly how the intensity changes. (2 marks)

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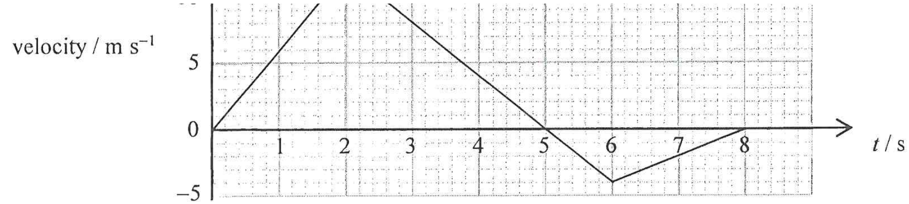

(i) Find the maximum height that the block can reach from $t = 0$ to 8 s. (2 marks)

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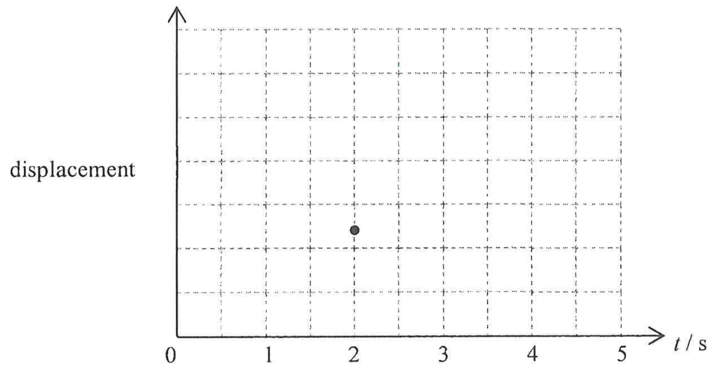
(ii) Sketch the displacement-time graph of the block from $t = 0$ to 5 s. The displacement at $t = 2$ s is marked. Labelling of values is not required and the displacement at $t = 0$ s is taken to be zero. (2 marks)



(iii) Find the tension of the string at $t = 5$ s. (3 marks)

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(i) Draw an arrow in the figure below to represent the direction of the acceleration of the block at that moment. (1 mark)

(ii) Find the horizontal displacement of the block from that moment until just before it reaches the ground. (3 marks)

(iii) Describe the energy conversion of the block from that moment until just before it reaches the ground. (2 marks)

(i) Write the nuclear equations corresponding to each of these two decay processes. (2 marks)

*(ii) The half-life of Bi-213 is 46 minutes. At time $t = 0$ s, its number of undecayed nuclei is 8.0×10^{11}. Find its activity (in Bq) at $t = 0$ s. (2 marks)

*(iii) Find the percentage decrease in the activity after 200 minutes. (2 marks)

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Figure 12.2(a) Figure 12.2(b)

(i) Compared to placing it in a badge worn on the chest, what is the advantage of placing the chip in the ring worn on the hand ? (1 mark)

(ii) The ring should be worn under the gloves. Explain briefly. (1 mark)

(iii) Which type(s) of radiation (α, β or γ) can reach the chip of the ring ? (1 mark)

END OF PAPER

Sources of materials used in this paper will be acknowledged in the *HKDSE Question Papers* booklet published by the Hong Kong Examinations and Assessment Authority at a later stage.

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